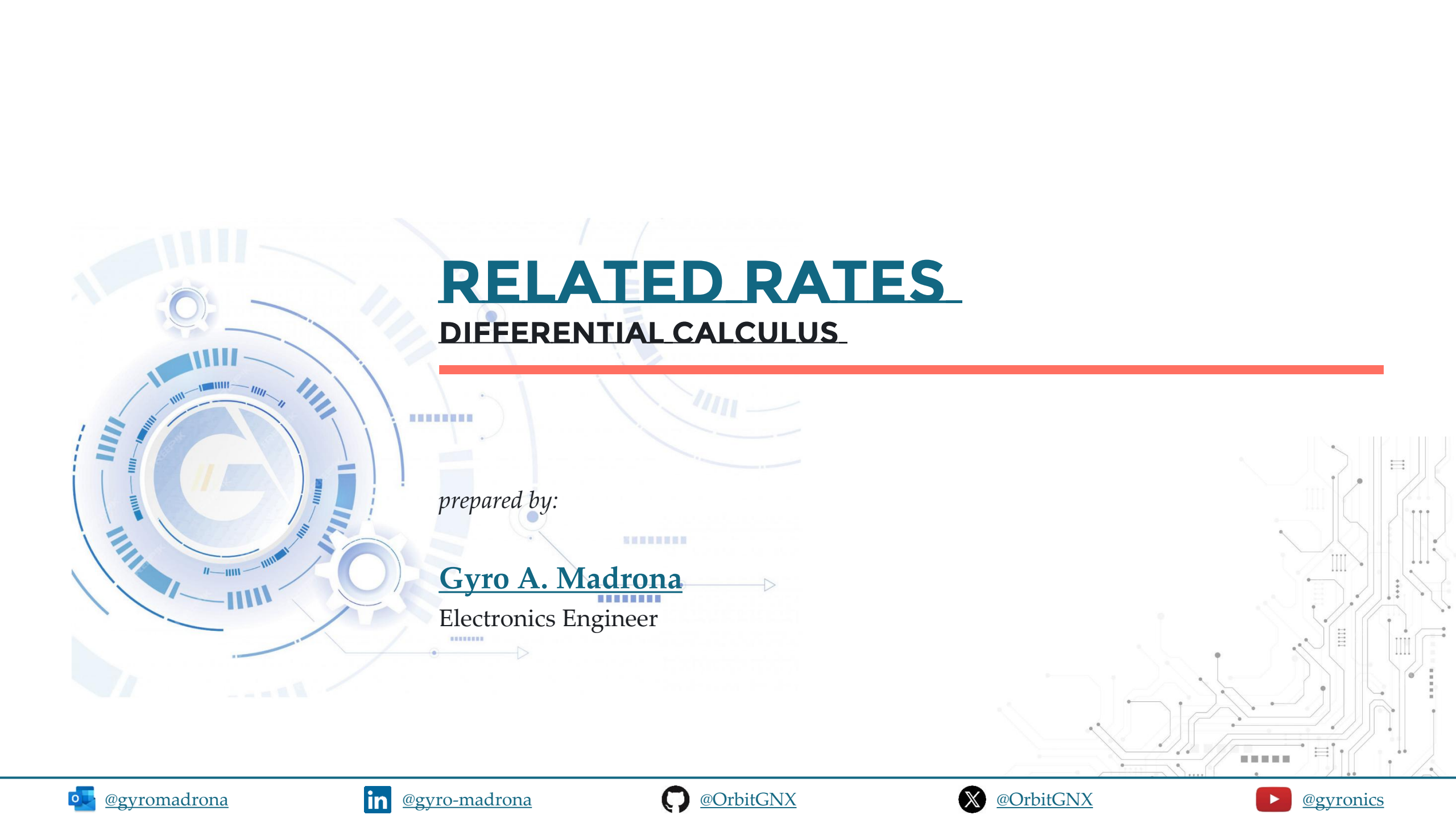




RELATED RATES

DIFFERENTIAL CALCULUS

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TOPIC OUTLINE

Related Rates Problem Solving



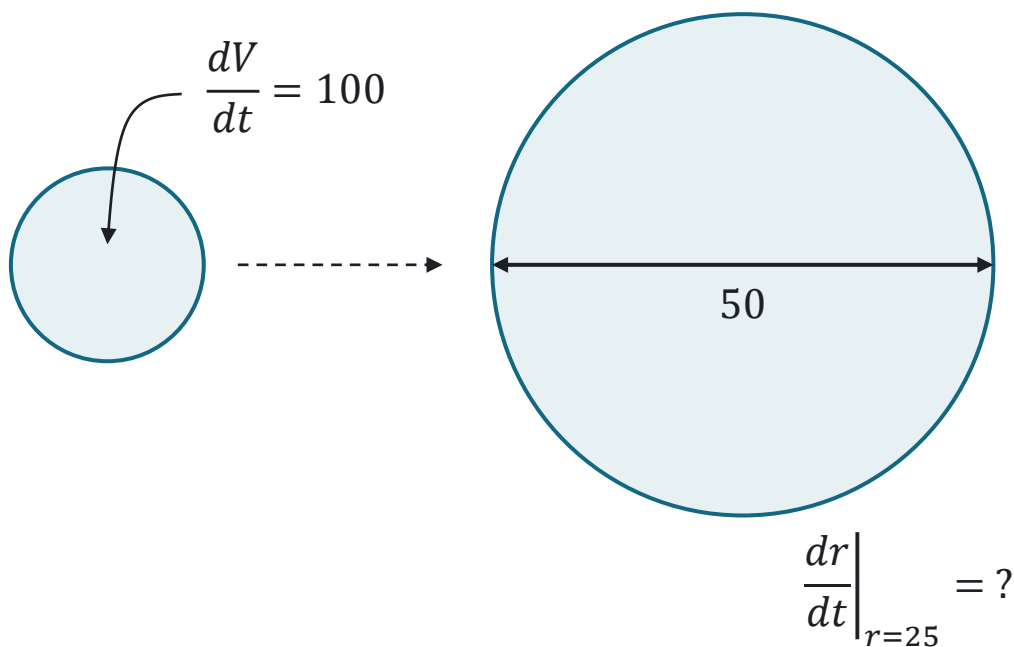
RELATED RATES PROBLEM

SOLVING



EXERCISE

Air is being pumped into a spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon increasing when the diameter is 50 cm ?



Solution

Volume of Sphere

$$V = \frac{4}{3} \pi r^3$$

$$\frac{d}{dt}(V) = \frac{4}{3} \pi \frac{d}{dt}(r^3)$$

$$\frac{dV}{dt} = \frac{4}{3} \pi (3r^2 \frac{dr}{dt})$$

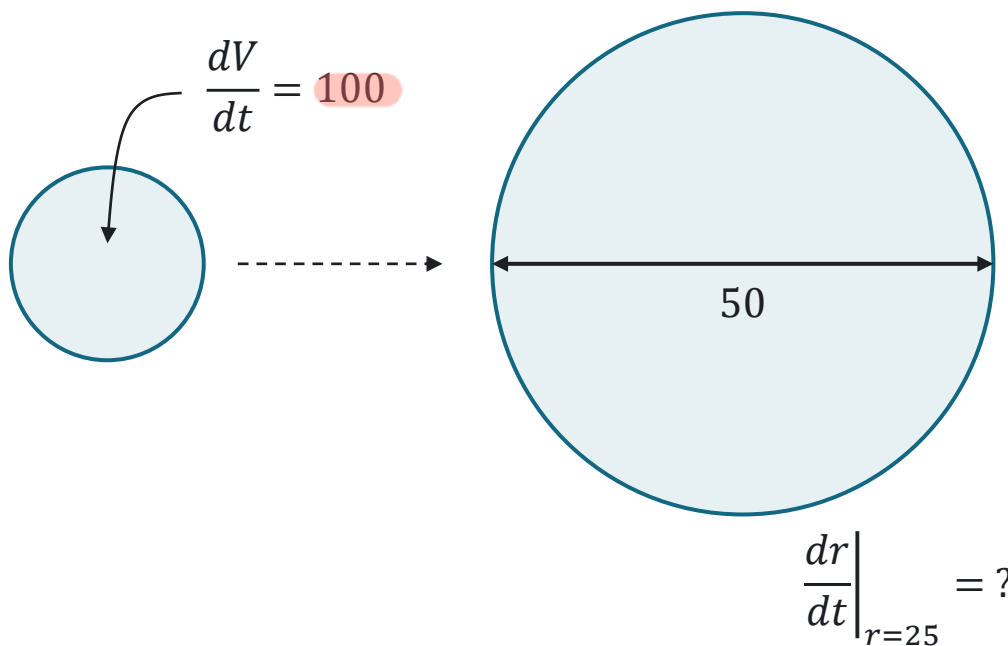
$$\frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$



EXERCISE

Air is being pumped into a spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon increasing when the diameter is 50 cm ?

$$r = 25$$



Solution

$$\frac{dr}{dt} = \frac{1}{4\pi r^2} \cdot \frac{dV}{dt}$$

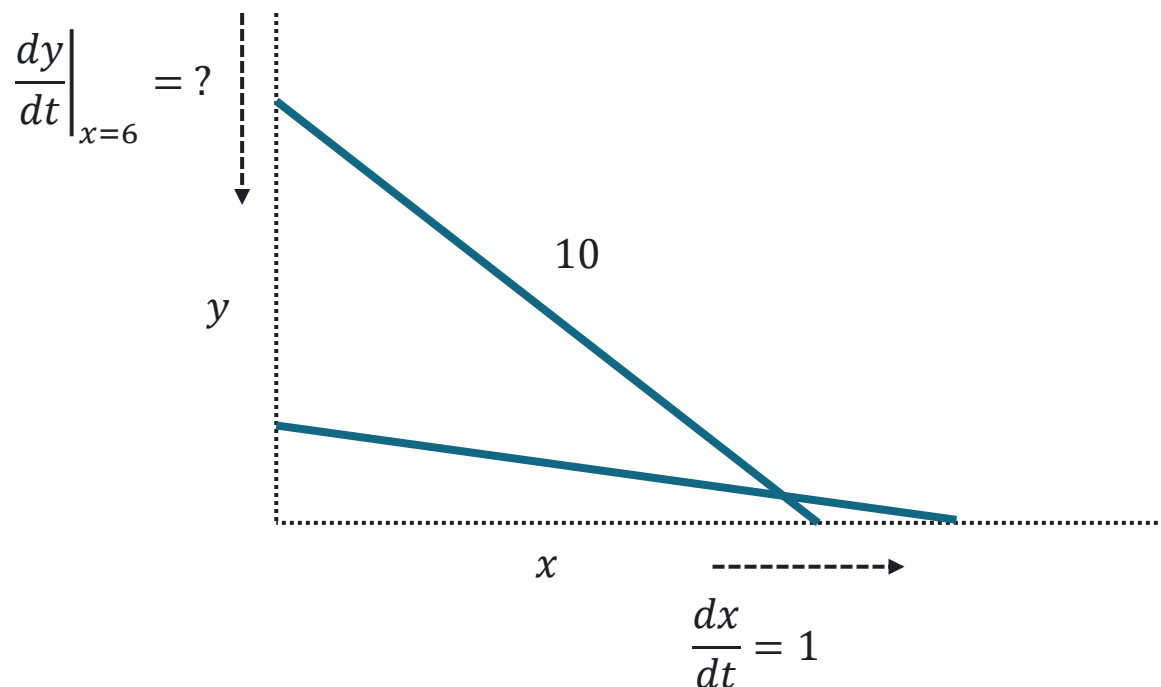
$$\left. \frac{dr}{dt} \right|_{r=25} = \frac{1}{4\pi (25)^2} \cdot (100)$$

$$\left. \frac{dr}{dt} \right|_{r=25} = \frac{1}{25\pi} \text{ cm/s}$$

ans

EXERCISE

A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 ft/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?



Solution

Pythagorean Theorem

$$x^2 + y^2 = 10^2$$

$$\frac{d}{dt}(x^2) + \frac{d}{dt}(y^2) = \frac{d}{dt}(10^2)$$

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

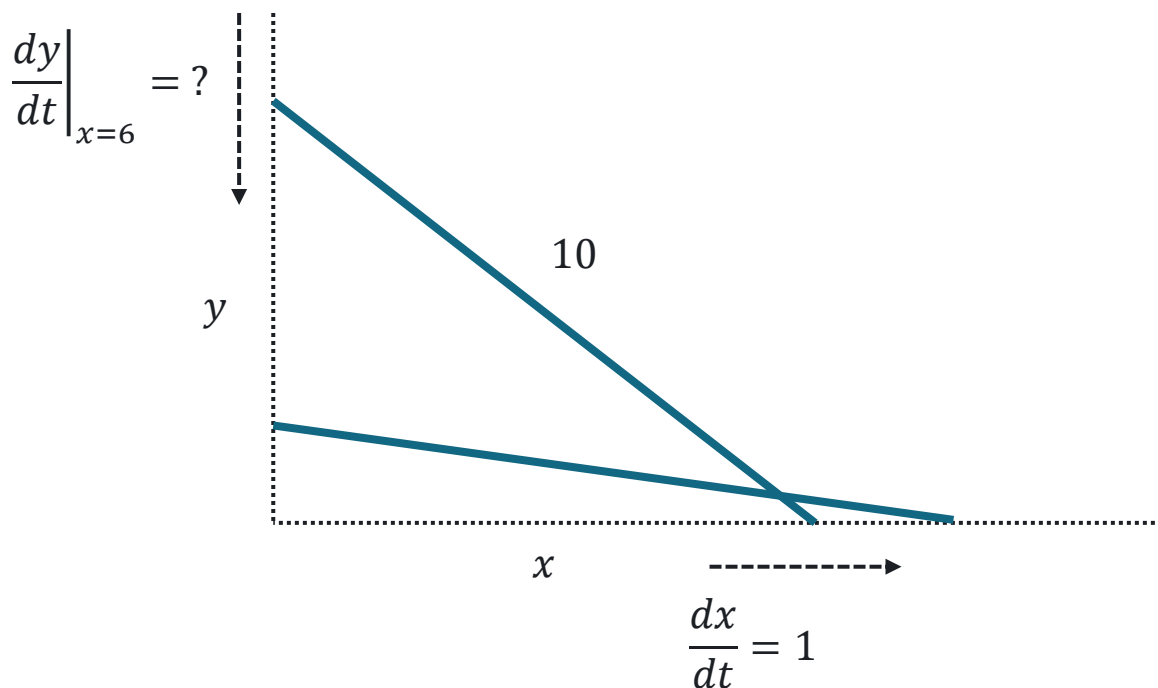
$$\cancel{2y} \frac{dy}{dt} = -\cancel{2x} \frac{dx}{dt}$$

$$\frac{dy}{dt} = -\frac{x}{y} \cdot \frac{dx}{dt}$$

EXERCISE

A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 ft/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?

$$x = 6$$



Solution

Pythagorean Theorem

$$x^2 + y^2 = 10^2$$

$$6^2 + y^2 = 100$$

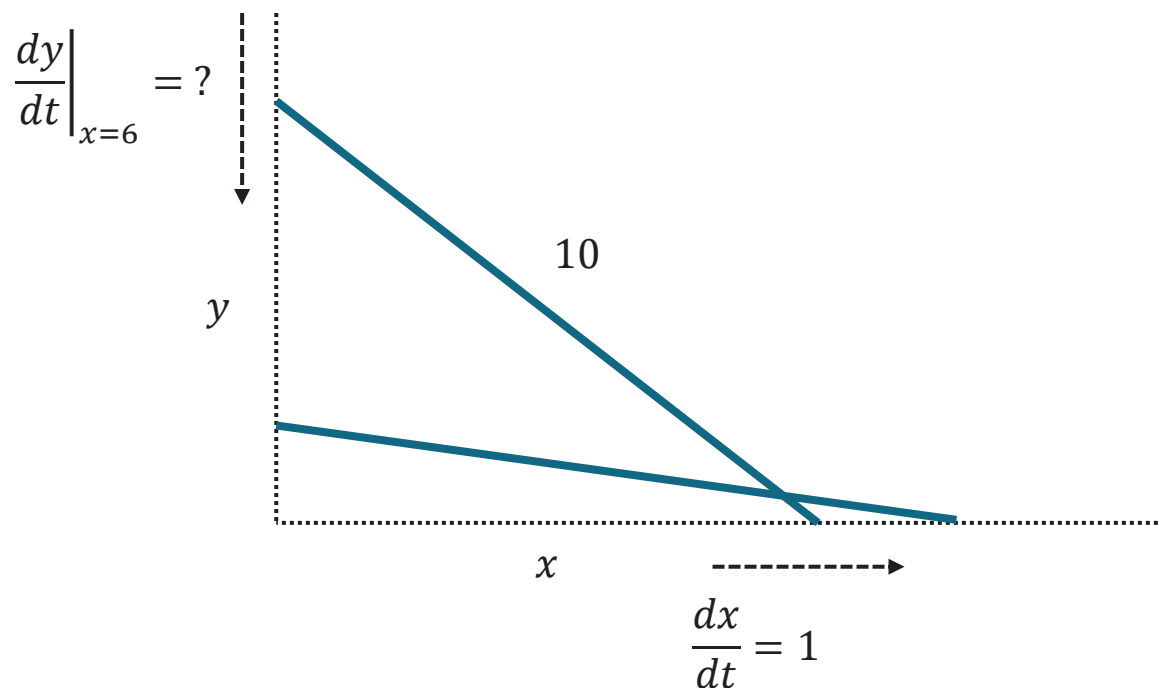
$$y^2 = 100 - 6^2$$

$$y = \sqrt{100 - 36}$$

$$\underline{y = 8}$$

EXERCISE

A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 ft/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?



Solution

$$\frac{dy}{dt} = -\frac{x}{y} \cdot \frac{dx}{dt}$$

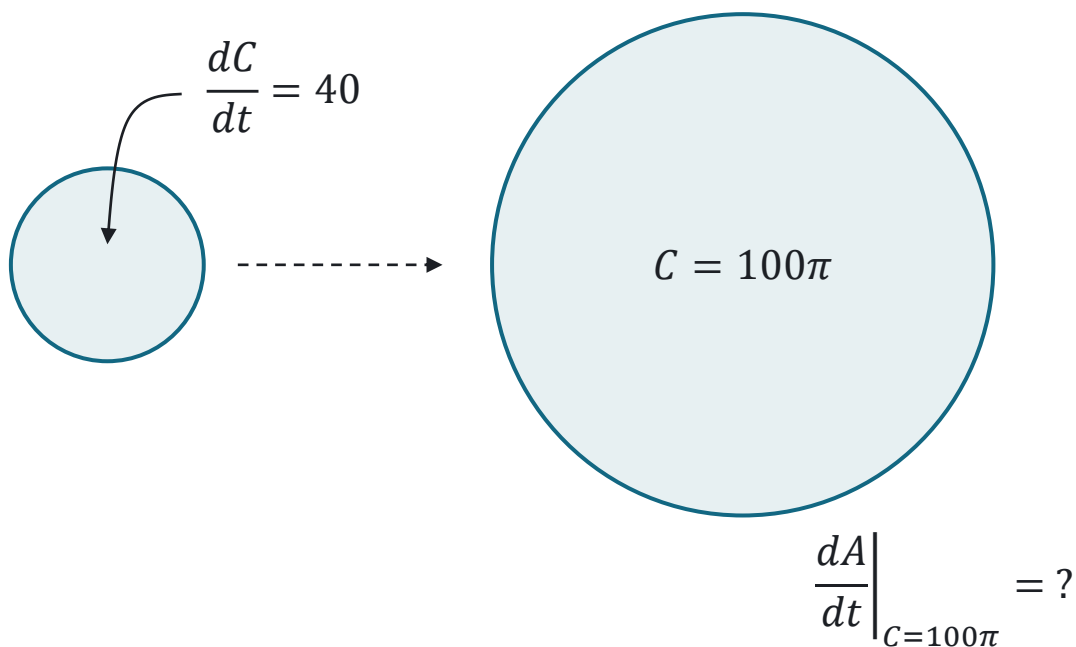
$$\frac{dy}{dt} \Big|_{y=8} = -\frac{6}{8} (1)$$

$$\frac{dy}{dt} \Big|_{y=8} = -\frac{3}{4} \text{ ft/s}$$

ans

EXERCISE

Oil spilled from a tanker spreads in a circle whose circumference increases at a rate of 40 ft/s . How fast is the area of the spill increasing when the circumference of the circle is $100\pi \text{ ft}$?



Solution

$$C = 2\pi r$$

$$r = \frac{C}{2\pi}$$

$$A = \pi r^2$$

$$A = \pi \left[\frac{C}{2\pi} \right]^2$$

$$A = \cancel{\pi} \frac{C^2}{\cancel{4\pi^2}}$$

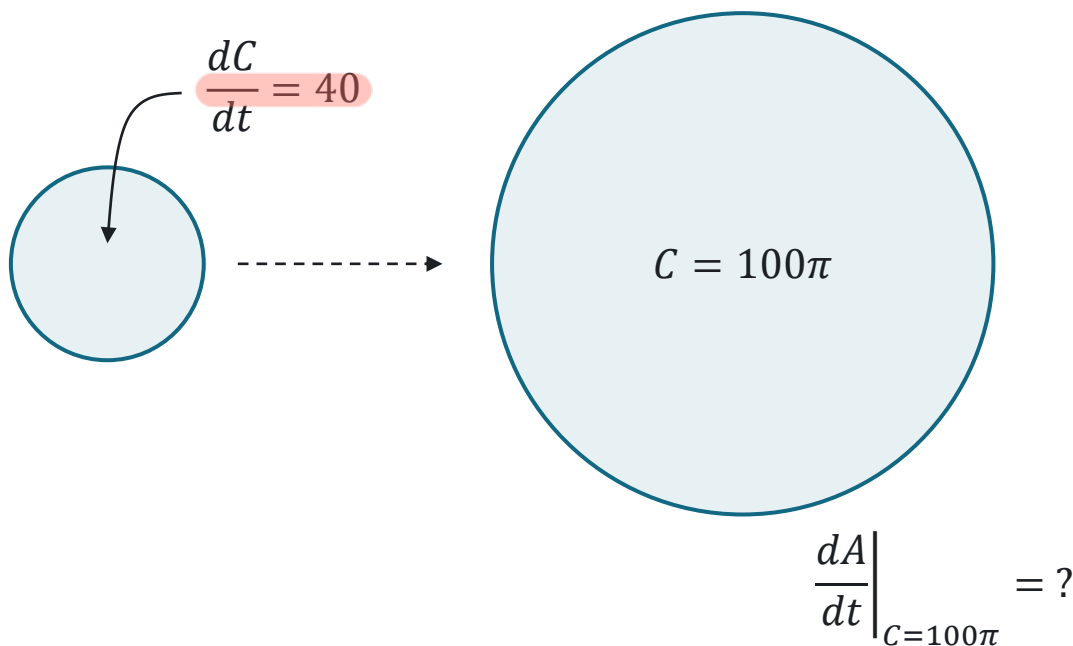
$$\underline{A = \frac{1}{4\pi} C^2}$$



EXERCISE

Oil spilled from a tanker spreads in a circle whose circumference increases at a rate of 40 ft/s . How fast is the area of the spill increasing when the circumference of the circle is $100\pi \text{ ft}$?

$$C = 100\pi$$



Solution

$$\frac{d}{dt}(A) = \frac{1}{4\pi} \frac{d}{dt}(C^2)$$

$$\frac{dA}{dt} = \frac{1}{4\pi} \cdot 2C \cdot \frac{dC}{dt}$$

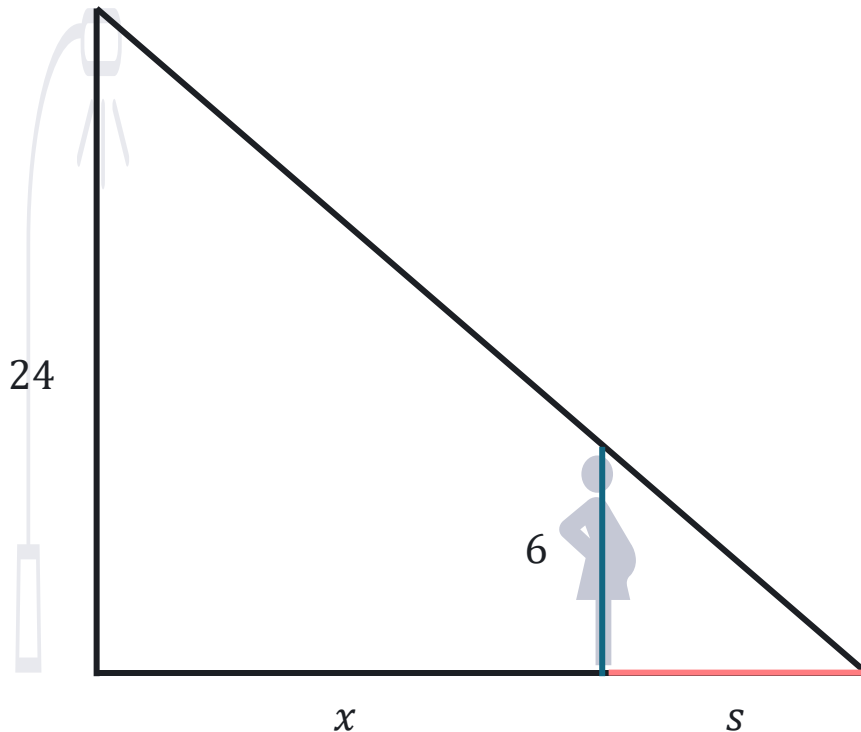
$$\left. \frac{dA}{dt} \right|_{C=100\pi} = \frac{1}{4\pi} (100\pi) \cdot (40)$$

$$\left. \frac{dA}{dt} \right|_{C=100\pi} = 2000 \text{ ft}^2/\text{s}$$

ans

EXERCISE

A 6-ft tall woman is walking at the rate of 4 ft/s away from a streetlamp that is 24 ft tall. How fast is the length of her shadow changing?



Solution

Similar triangles

$$\frac{24}{x+s} = \frac{6}{s}$$

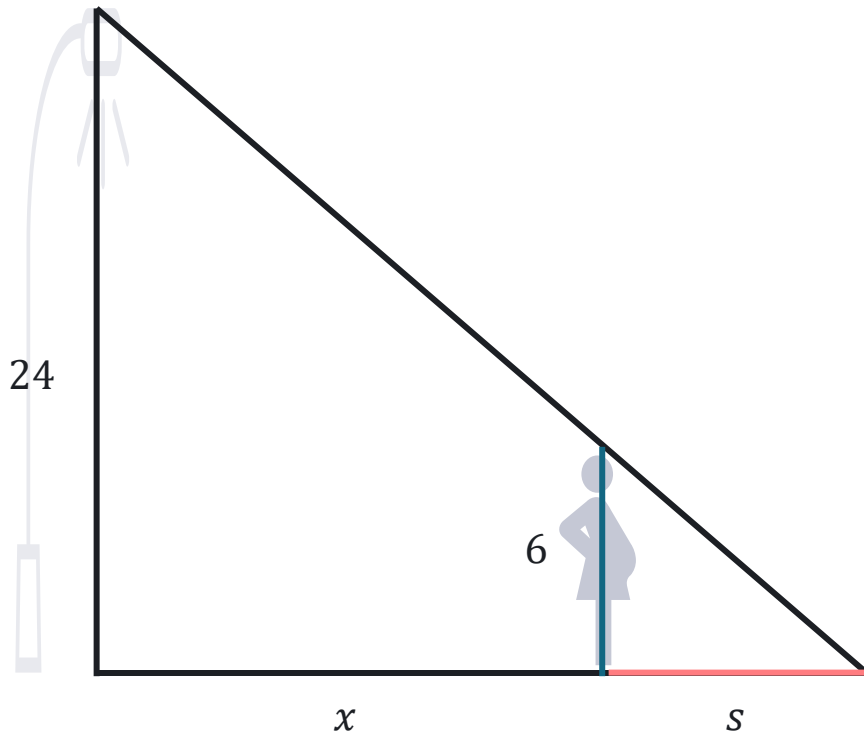
$$\begin{array}{rcl} 24s & = & 6x + 6s \\ \underline{-6s} & & \underline{-6s} \end{array}$$

$$\frac{18s}{18} = \frac{6x}{18}$$

$$s = \frac{1}{3}x$$

EXERCISE

A 6-ft tall woman is walking at the rate of 4 ft/s $\frac{dx}{dt}$ away from a streetlamp that is 24 ft tall. How fast is the length of her shadow changing?



Solution

$$\frac{ds}{dt} = \frac{1}{3} \cdot \frac{dx}{dt} \rightarrow 4$$

$$\boxed{\frac{ds}{dt} = \frac{4}{3} \text{ ft/s}}$$

ans,

EXERCISE

The voltage V , in electrical circuit is related to the current I , and the resistance R , by the equation $V = IR$. The current is decreasing at -4 ^{amps/sec} as the resistance increases at 20 ^{ohms/sec}. How fast is the voltage changing when the voltage is 100 volts and the current is 20 amps?

$$I = 20$$

$$R = \frac{V}{I}$$

$$R = \frac{100}{20}$$

$$R = 5 \Omega$$

Solution

$$V = IR$$

$$\frac{d}{dt}(V) = \frac{d}{dt}(IR)$$

$$\frac{dV}{dt} = I \frac{dR}{dt} + R \frac{dI}{dt}$$

$$\left. \frac{dV}{dt} \right|_{\substack{I=20 \\ R=5}} = 20(20) + 5(-4)$$

$$\left. \frac{dV}{dt} \right|_{\substack{I=20 \\ R=5}} = 380 \text{ V/s}$$

ans

EXERCISE

If a snowball melts so that its surface area decreases at a rate of $1 \text{ cm}^2/\text{min}$, find the rate at which the diameter decreases when the diameter is 10 cm .

$$\frac{dD}{dt}$$

$$D = 10$$

Surface area of sphere

$$A = 4\pi r^2$$

$$A = 4\pi \left(\frac{D}{2}\right)^2$$

$$A = \cancel{4}\pi \frac{D^2}{\cancel{4}}$$

$$\underline{A = \pi D^2}$$

Solution

$$\frac{d}{dt}(A) = \pi \frac{d}{dt}(D^2)$$

$$\frac{dA}{dt} = \pi \left(2D \cdot \frac{dD}{dt} \right)$$

$$\frac{dD}{dt} = \frac{1}{2\pi D} \cdot \frac{dA}{dt} \rightarrow 1$$

$$\left. \frac{dD}{dt} \right|_{D=10} = \frac{1}{2\pi(10)}$$

$$\left. \frac{dD}{dt} \right|_{D=10} = \frac{1}{20\pi} \text{ cm/s}$$

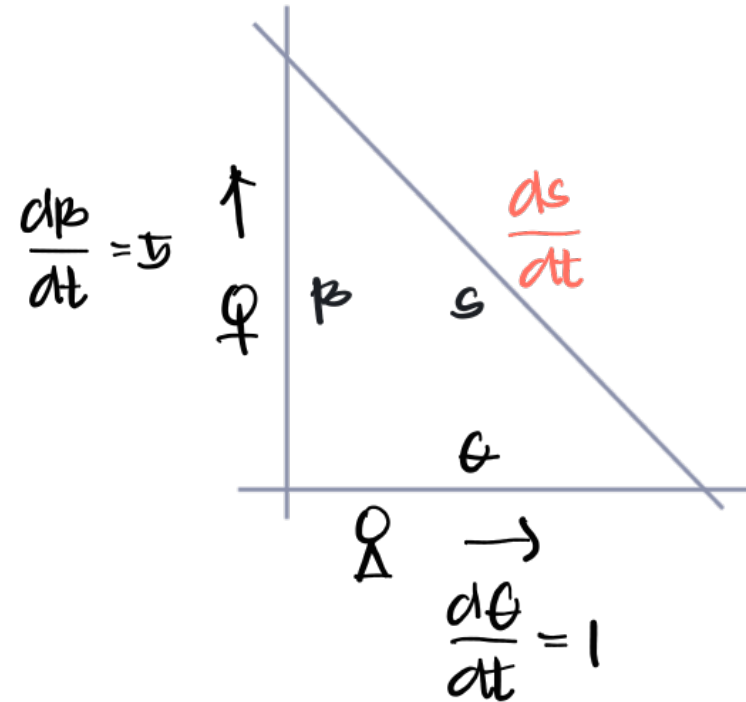
ans



EXERCISE

On a rainy day, a girl broke up with her boyfriend after being together for eight long years. They decided to separate at the place where everything about them began, at the same time. The boy is due north crying and running at a rate of 5 ft/s and the girl is walking due east at the rate of 1 ft/s thinking if she made the right decision. **How fast** ^{ds/dt} are they separating from each other ^{$t=2$} 2 seconds after they started moving to a new life without each other?

Solution



Pythagorean theorem

$$s^2 = b^2 + a^2$$



EXERCISE

On a rainy day, a girl broke up with her boyfriend after being together for eight long years. They decided to separate at the place where everything about them began, at the same time. The boy is due north crying and running at a rate of 5 ft/s and the girl is walking due east at the rate of 1 ft/s thinking if she made the right decision. **How fast** ^{ds/dt} are they separating from each other ^{$t=2$} 2 seconds after they started moving to a new life without each other?

Solution

$$\frac{d}{dt}(s^2) = \frac{d}{dt}(b^2) + \frac{d}{dt}(g^2)$$

$$2s \frac{ds}{dt} = 2b \frac{db}{dt} + 2g \frac{dg}{dt}$$

$$\text{let } t = 2$$

$$b = 2\cancel{s} \cdot \frac{5\text{ft}}{\cancel{s}} \quad g = 2\cancel{s} \cdot \frac{1\text{ft}}{\cancel{s}}$$

$$b = 10 \text{ ft} \quad g = 2 \text{ ft}$$

$$s = \sqrt{10^2 + 2^2}$$

$$s = 10.20 \text{ ft}$$



EXERCISE

On a rainy day, a girl broke up with her boyfriend after being together for eight long years. They decided to separate at the place where everything about them began, at the same time. The boy is due north crying and running at a rate of 5 ft/s and the girl is walking due east at the rate of 1 ft/s thinking if she made the right decision. **How fast** ^{ds/dt} are they separating from each other ^{$t=2$} 2 seconds after they started moving to a new life without each other?

Solution

$$2(\cancel{10.20}) \frac{ds}{dt} = 2(\cancel{10}) \overset{\frac{db}{dt}}{(\cancel{5})} + 2(\cancel{2}) \overset{\frac{dg}{dt}}{(1)}$$

$$\cancel{20.4} \frac{ds}{dt} = 104$$

$$\frac{ds}{dt} = \frac{104}{20.4}$$

$$\boxed{\frac{ds}{dt} \Big|_{t=2} = 5.1 \text{ ft/s}}$$

ans



LABORATORY

